

Fahad Ibne Rahman

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BIOGRAPHY

I am an M.Sc. student in Biomedical Engineering at Islamic University, Bangladesh, and a Research Assistant at the Bio-Imaging Research Laboratory with over three years of research experience in medical image processing, deep learning, and AI for healthcare. My research has resulted in three peer-reviewed IEEE conference publications, and I have contributed as a peer reviewer for four manuscripts submitted to peer-reviewed journals. My current work focuses on developing novel activation functions and Segmentation architectures for Medical Imaging.

RESEARCH INTERESTS

Deep Learning · Medical Image Analysis · Computer Vision · Machine Learning · AI in Healthcare · Optimization Algorithms · Bio-signal Processing.

EDUCATION

M.Sc. (Engineering) – Islamic University, Bangladesh <i>Biomedical Engineering; Ongoing</i>	Kushtia-7003, Bangladesh Nov 2025 – Present
B.Sc. (Engineering) – Islamic University, Bangladesh <i>Biomedical Engineering; CGPA: 3.79/4.00 (2nd of 50 students)</i>	Kushtia-7003, Bangladesh Jan 2020 – Aug 2025
Milestone College Uttara <i>H.S.C.; Science; GPA: 5.00/5.00</i>	Dhaka-1230, Bangladesh Jun 2016 – Jul 2018
Jamalpur R.M. Bidyapith <i>S.S.C.; Science; GPA: 4.94/5.00</i>	Kaligonj, Gazipur, Bangladesh Jan 2011 – Mar 2016

RELEVANT COURSEWORK

Medical Imaging · Neural Engineering · Bio-Signal Processing · Biomedical Instrumentation and Measurement · Biomedical Microsystems · Bioinformatics · Machine Learning in Healthcare · Biosensors and Bioelectronics · Drug Delivery Systems.

PUBLICATIONS

Conference Proceedings

- Fahad Ibne Rahman**, Nusrat Islam, Md Emamul Hossen, Md. Khairul Islam, “A Deep Learning Approach Based on XAI and ViT-GRU Hybrid Model for Brain Tumor Classification Using MRI Images,” *IEEE 1st International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN 2025)*, *IEEE Xplore*.
DOI: [10.1109/QPAIN66474.2025.11171653](https://doi.org/10.1109/QPAIN66474.2025.11171653)
- Fahad Ibne Rahman**, Nusrat Islam, Md. Khairul Islam, “FocalNet: An Attention-Free Novel Architecture for Brain Stroke Classification Using CT Imaging,” *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN 2026)*, *IEEE Xplore*.
DOI: [10.1109/QPAIN69676.2026.11545628](https://doi.org/10.1109/QPAIN69676.2026.11545628)
- Nusrat Islam, **Fahad Ibne Rahman**, Suman Biswas, “SNM-TransNet: A Novel SNMish Activation-Based Attention-Enhanced Hybrid Architecture for Accurate Breast Cancer Image Classification,” *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN 2026)*, *IEEE Xplore*.
DOI: [10.1109/QPAIN69676.2026.11546422](https://doi.org/10.1109/QPAIN69676.2026.11546422)

MANUSCRIPTS UNDER REVIEW

Journal Papers

- Fahad Ibne Rahman**, Nayan Kumar Sarkar, Nusrat Islam, Md. Ripaj Uddin, “Discovery of Wee1 Kinase Inhibitors from *Mangifera indica* Leaf Phytochemicals Ellagic Acid, Vitexin, and Isoswertisin for Colon Cancer: An Integrated Computational Approach,” – *Phytomedicine (Elsevier)*.

MANUSCRIPTS IN PREPARATION

- Fahad Ibne Rahman**, Nusrat Islam, “STanh: A Novel Self-Gated Activation Function for Robust and Efficient Deep Learning Across Convolutional and Transformer Architectures,” – *Manuscript in Preparation*.
- Fahad Ibne Rahman**, Nusrat Islam, “FPVT-UNet: A Multi-Scale Parallel Focal Modulation-PVTv2 Bottleneck Architecture for Automated Brain Stroke Lesion Segmentation from NCCT Images,” – *Manuscript in Preparation*.
- Nusrat Islam**, Fahad Ibne Rahman, “PVTv2-DVRNet: A Dynamic Vessel Refinement Network for Accurate Retinal Vessel Segmentation in Fundus Images,” – *Manuscript in Preparation*.

RESEARCH EXPERIENCE

- **Bio-Imaging Research Lab** Islamic University, Bangladesh
Research Assistant May 2023 – Present
 - * **Director:** Dr. Md. Khairul Islam, Associate Professor, Department of Biomedical Engineering, Islamic University, Bangladesh.
 - * **Research Area:** Medical image processing, computer vision, deep learning, signal processing, and health informatics.
- **Sono Hospital Limited** Kushtia, Bangladesh
Clinical Fieldwork May 2–4, 2024
 - * **Supervisor:** Dr. Md. Khairul Islam, Associate Professor, Department of Biomedical Engineering, Islamic University, Bangladesh.
 - * **About:** Hands-on clinical exposure to major medical imaging modalities at Sono Hospital Ltd., observing and operating CT scan, MRI, and X-ray equipment and their role in diagnostic imaging workflows.

UNDERGRADUATE PROJECT

VSL-Arc: A Novel Hybrid Deep Learning Architecture for Accurate Lung Cancer Classification Using CT Images
Supervisor: Dr. Md. Khairul Islam, Associate Professor, Department of Biomedical Engineering, Islamic University, Bangladesh

HONORS AND AWARDS

1. **University Merit Scholarship** (B.Sc.) – 2023 & 2024
2. **University General Scholarship** (B.Sc.) – 2022
3. **Best Abstract Award** – “XAI-Driven ViT-GRU Deep Learning Framework for MRI-Based Brain Tumor Classification,” *IEEE CS BDC Summer Symposium 2025, HSTU, Dinajpur, Bangladesh*. (Certificate). – 2025
4. **Best Presenter Award** – “FocalNet: An Attention-Free Novel Architecture for Brain Stroke Classification Using CT Imaging,” *IEEE QPAIN 2026, CUET, Chattogram, Bangladesh*. – 2026
5. **Best Presenter Award** – “SNM-TransNet: A Novel SNMish Activation-Based Attention-Enhanced Hybrid Architecture for Accurate Breast Cancer Image Classification,” *IEEE QPAIN 2026, CUET, Chattogram, Bangladesh*. – 2026

JOURNAL REVIEWS

1. Reviewer – Computational Biology and Chemistry; Publisher: Elsevier; reviewed 4 papers (Certificate).

TECHNICAL SKILLS

Languages: Python

Frameworks/Libraries: TensorFlow, Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

Tools & Platforms: GitHub, Kaggle, LaTeX (Overleaf), MS Office (Word, PowerPoint, Excel)

Soft Skills: Leadership, Project Management, Time Management, Technical Writing, Public Speaking

LEADERSHIP AND VOLUNTEERING

- **General Secretary – Public University Students Association of Kaliganj** Apr 2025 – Present
 - * Represents students from public universities originating from Kaliganj Upazila, Gazipur, Bangladesh. Promotes academic collaboration, mutual support, and student development. Organizes scholarship programs for school and college students and conducts university admission-related sessions, seminars, and awareness programs to guide prospective students in higher education.
- **Executive Member – Gazipur Students Welfare Association** Mar 2023 – Aug 2025
 - * Non-political welfare association for Islamic University, Bangladesh students from Gazipur district

REFEREES

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